PRIYANKA GONUGUNTA

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PROFESSIONAL SUMMARY

Senior Data Scientist and Machine Learning Engineer with 10+ years of experience delivering predictive analytics, recommendation systems, scalable data engineering, and cloud-based ML solutions across financial services and retail. Strong background in Python, SQL, AWS SageMaker, Snowflake/Snowpark, MLOps, model training/deployment, CI/CD, unit testing, and production migration from SAS/Netezza. Hands-on independent Generative AI development using sentence-transformer embeddings, vector similarity, retrieval-augmented generation (RAG), Ollama/Llama 3.2, and grounded responses with source references. Exploring Agentic AI architecture, tool routing, orchestration, memory/state, and MCP concepts; differentiates prototype experience from production delivery.

TECHNICAL SKILLS

Programming & Data Python, SQL, SAS, R, Pandas, PySpark, Apache Spark, Snowflake, Snowpark, Netezza, NoSQL, data modeling, ETL/ELT

Machine Learning Regression, classification, clustering (K-means), RFM, recommendations, time series, feature engineering, EDA, A/B testing, hypothesis testing, SMOTE, AUROC; TensorFlow, PyTorch, NLP, computer vision, BERT, RoBERTa

Generative AI / RAG LLM applications, semantic search, Sentence Transformers, text chunking, 384-dimensional embeddings, vector representations, cosine similarity, top-k retrieval, prompt engineering, context grounding, citation review, hallucination checks; Ollama, Llama 3.2, Streamlit

Agentic AI (Prototype / Learning) Deterministic tool routing, retrieval/calculator tool workflows, agent orchestration principles, tool calling concepts, LangGraph and Model Context Protocol (MCP) fundamentals; not claimed as production implementations

Cloud / MLOps / Tools AWS SageMaker, S3, IAM, EC2, Redshift, MLflow, Docker, FastAPI, Git/GitHub, CI/CD, YAML, pytest, Amazon Bedrock (listed skills); Power BI, Tableau, MicroStrategy, JIRA, Confluence, VS Code

PROFESSIONAL EXPERIENCE

Toyota Financial Services | Data Scientist / Analyst | Nov 2025 – Oct 2026

- Modernized SAS/Netezza-based financial analytics into Python and Snowflake/EDP production workflows, improving scalability, modularity, automation, and maintainability of business-critical pipelines.
- Enhanced the Post-Booking Cashflow Engine for large retail and lease portfolios; implemented Python preprocessing, cashflow calculations, loss-curve integration, and reporting for downstream ALM/Treasury analysis.
- Reworked loss-curve processing with vectorized Snowflake SQL, eliminating unnecessary intermediate datasets and Python pivots while lowering data-loading, memory, and processing overhead.
- Migrated and validated ALM/Treasury curve-generation and forecasting workflows; reconciled legacy vs. Snowflake outputs and investigated variances across LTV, FICO, MOB, vintage, curve mapping, and portfolio segments.
- Built pytest unit and regression suites to verify source-to-target parity, complex financial business rules, and calculation accuracy; improved the repeatability of production QA and release validation.
- Supported SageMaker-related ML/analytical workflows, Git branching, CI/CD conventions, YAML configuration, technical documentation, and cross-functional delivery with Data Science, Finance, Treasury, and Engineering.
- Applied Generative AI assistants (ChatGPT and Claude) to accelerate SAS-to-Python and SQL conversion, legacy-code interpretation, debugging, refactoring, unit-test scaffolding, and technical documentation; verified AI-generated suggestions through code review, pytest regression checks, and reconciliation against legacy results before adoption.

Toyota Financial Services | Machine Learning Engineer | Apr 2025 – Oct 2025

- Migrated 30,000+ lines of legacy SAS into optimized, modular Python/Snowflake pipelines; reported 40% less code complexity and over 70% reduction in manual intervention through reusable automation.
- Integrated AWS SageMaker training/deployment workflows and automation across the ML lifecycle, contributing to a reported 60% improvement in retraining efficiency.
- Developed unit and regression validation frameworks with approximately 99% parity to legacy SAS results, reducing QA turnaround from days to hours.
- Built YAML-driven environment configuration and Git/CI/CD-friendly deployment practices; reported 50% faster deployments and 30–40% faster large-scale data transformations using Snowpark/optimized SQL.

Landmark Group | Senior Data Scientist | Nov 2018 – Dec 2022

- Led six analysts delivering customer intelligence, campaign analytics, segmentation, predictive modeling, and measurable sales/category performance improvements in a large retail environment.
- Developed K-means/RFM segmentation and predictive marketing strategies, reporting AED 150M incremental revenue, 30% higher campaign effectiveness, and 20% improvement in campaign ROI.
- Built image-processing and similarity-based product recommendation models for Splash Apparel, increasing product engagement by 25%; recognized for the color-based recommendation algorithm.
- Designed Python/TensorFlow/AWS SageMaker predictive workflows and scalable Spark processing; partnered with Sales, Marketing, Category, and leadership teams to turn ML findings into commercial actions.

Sybrant Technologies | Senior Data Analyst | Aug 2015 – Dec 2017

- Developed SQL/Python reporting, SAS macro automation, and Power BI/Tableau dashboards for operational, financial, and risk stakeholders.
- Built loan-default and customer-churn prediction with logistic regression/decision trees; performed segmentation, imbalance handling (SMOTE), testing, and risk analysis to support faster decision-making.

Botcode Integration | Data Analyst | Aug 2012 – Jul 2015

- Cleaned and integrated multi-source SQL/Python datasets; automated recurring extracts, dashboards, KPI reporting, exploratory analysis, and customer-segmentation reporting.

INDEPENDENT GENERATIVE AI & AGENTIC AI PROJECTS

Financial Document Semantic Search | Python, Sentence Transformers, NumPy, Streamlit

- Created a local semantic retrieval pipeline over four synthetic financial documents; split text into chunks, embedded eight passages into 384-dimensional vectors, and ranked results via cosine similarity and top-k search.
- Built an interactive interface showing retrieved passages, similarity scores, source files, and execution traces to explain embedding/vector behavior and retrieval decisions.

Financial Document RAG Assistant | Ollama, Llama 3.2, Python, Streamlit

- Extended semantic search into document-only RAG: retrieved passages are incorporated into the LLM prompt to generate context-grounded responses with source references.
- Tested unsupported-question behavior and demonstrated safe abstention when the supplied documents lacked requested 2026 loan-default statistics; compared retrieval-only vs. generated responses.

Diwali AI Celebration Studio | GenAI, RAG, Streamlit

- Developed a themed interactive application combining Diwali-document semantic retrieval, LLM-assisted personalized greeting generation, and an event budget calculator.
- Prepared a public-hosting version with customized visuals, mobile-friendly presentation, and optional hosted LLM integration; kept local Ollama development separate from public inference.

Agentic AI Foundations | Independent Prototyping

- Implemented deterministic request-to-tool routing in mini-agent prototypes and studied how LLM-driven tool selection, agent state, orchestration loops, LangGraph, and MCP can extend RAG workflows. Production autonomous-agent implementation is a future development goal.

EDUCATION

M.S. Data Science — Webster University (Dec 2024) | M.S. Business Analytics — Praxis Business School (2019) | M.S. Information Technology — Anna University (2012)

AWARDS & RECOGNITION

Landmark Saviour Award; Exceptional Support to CI GCC; Ramadan Dashboard Spot Award; recognition for Splash Apparel color-based recommendation algorithm; Anna University IT department rank 7.